2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) | 979-8-3315-8618-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/EMBC58623.2025.11254113

A Multi-Band Self-Attention Network for Motor Imagery Classification

Qingjun Song. Author[1] and Guixia Kang. Author[2,*]

Abstract— **Brain-computer interface (BCI) systems create a novel communication method between humans and machines by translating human thoughts into actionable commands to control external devices. Motor imagery (MI) electroencephalogram (EEG) signals have significant applicability in various medical and non-medical industries, including stroke rehabilitation, wheelchair control, and drone operation. However, the practical application of EEG remains limited by the decoding performance and generalization ability of MI signals.**

This study introduces a multi-branch self-attention network for motor imagery (MI) signal classification. Each branch independently processes EEG signals decomposed into distinct frequency bands through convolutional neural networks (CNNs) and multi-head self-attention (MHA) mechanisms, enabling the extraction of both fundamental and discriminative spatial-temporal features. To further capture dynamic temporal dependencies, long short-term memory (LSTM) networks are integrated. We systematically evaluate three signal decomposition ensemble empirical mode decomposition (EEMD), wavelet packet decomposition (WPD), and brain rhythm-based decomposition—to optimize feature representation. Extensive experiments on the BCI Competition IV 2a dataset demonstrate state-of-the-art performance, with subject-dependent and subject-independent accuracies of 84.04% and 71.67%, respectively. Comparative analyses against benchmark models (EEGNet, EEGTCNet, ShallowConvNet, etc.) validate the superiority of our approach in classification accuracy and generalization capability.

Clinical relevance— **This study investigates the methods for decoding motor imagery EEG signals and establishes the positive role of each module in classification. The improvement in accuracy can lead to better outcomes in medical applications such as controlling prosthetics, wheelchairs, and stroke rehabilitation.**

Key Words— **Self-attention, Convolutional Neural Networks, Motor Imagery, LSTM, Signal Frequency Decomposition**

I. INTRODUCTION

Brain-Computer Interface (BCI) systems provide a new way of human-computer communication by analyzing the subject's motor state through brain signals and transforming brain thoughts into real commands to control external devices [1]. In terms of brain signal acquisition, brain activity can be recorded using various technologies, such as Electroencephalography (EEG). EEG is a non-invasive method that records the electrical activity of the brain, capturing the brain signal as a two-dimensional matrix of real values (time and channels) on the scalp. EEG is widely used in various technologies and industrial applications due to its ease of use, low cost, low risk, portability, and high temporal resolution [2] .

Motor Imagery (MI) refers to the process where subjects mentally visualize moving a part of their body, such as the left or right hand, both feet, or the tongue. By decoding the brainwave signals generated during motor imagery, it is possible to analyze the corresponding imagined movements of the subject. This technique has broad applications in various medical fields, such as controlling prosthetic limbs and wheelchairs, as well as aiding in stroke rehabilitation [3]. However, the decoding of motor imagery signals is not yet sufficiently accurate, and several challenges remain.

Current research on BCI based on motor imagery (MI) faces several key issues: (1) Low signal-to-noise ratio (SNR): Most BCIs are based on EEG recorded using non-invasive electrodes placed on the scalp to minimize harm to the subject. In this case, factors such as the condition of hair and scalp, as well as the material and quality of the electrodes, can affect the conductivity between the electrode and the scalp, leading to inaccuracies in signal capture, which is one of the main causes of low SNR. Additionally, small head movements during EEG signal collection can lead to changes in electrode-skin contact, further contributing to low SNR. (2) Non-stationarity of recorded EEG signals: External environmental changes or internal bodily conditions, such as noise, attention, and fatigue, can unpredictably impact EEG signals [4] . These signals may mix with other noise, making signal extraction more challenging. (3) Individual differences among subjects: EEG signals from different individuals are as unique as fingerprints [5]. When a person imagines a movement, the uniqueness of their EEG signal is more pronounced, and therefore, predictions can only be made using that individual's own EEG data, which makes motor imagery more challenging to apply. Addressing these issues is crucial for advancing the practical application of EEG signals, not only for improving convenience in daily life but also for making significant contributions to the field of neuroscience.

To tackle these problems, many approaches have been proposed, primarily based on manually extracted features and deep learning methods, which extract features in the frequency, time, and spatial domains for classification or use end-to-end networks for direct classification. Common Spatial Pattern (CSP) is a classical signal processing method for addressing low SNR, which can extract useful feature signals from raw EEG data [6]. In recent years, the use

*This work was supported by the State Key Program of the National Natural Science Foundation of China (82030037)

[1]Qingjun Song is with School of Information and Communication Engineering, Communication Engineering, Beijing University of Posts and Telecommunications, Beijing, China (e-mail: 93982647@qq.com)

[2]Guixia Kang is with Key Lab of Ubiquitous Network of Ministry of Education, Beijing University of Posts and Telecommunications, Beijing, China (corresponding author to provide e-mail: gxkang@bupt.edu.cn)

of traditional classifiers [7], [8], [9], [10] or neural network classifiers [11], [12], [13], [14] has expanded CSP methods (including FBCSP), enhancing the ability to extract useful information from recorded EEG signals and improving classification accuracy for MI tasks. The number of studies based on deep learning methods has also been rapidly increasing. Compared to traditional machine learning, deep learning can learn significant features from EEG data without requiring preprocessing and strong feature extraction. Recent years have seen different deep learning architectures proposed for MI classification, including Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Graph Convolutional Networks. In 2018, EEGNet [15], a deep learning model specifically designed for EEG data processing, was introduced. It uses a compact structure with convolutional and pooling layers to capture the spatiotemporal features in EEG signals. This model can efficiently extract relevant information from brain signals without requiring significant preprocessing or feature engineering, making it widely applicable in BCI and neuroscience. Subsequently, a variant of CNN, the Temporal Convolutional Network (TCN) [16], was proposed, and later, Ingolfsson et al. [17] introduced the EEG-TCN model, which combined TCN with the well-known EEGNet architecture, achieving high accuracy on the BCI competition IV 2a dataset [18]. Scholars from the International Neuroscience Institute proposed an EEGNet-LSTM classification method that combined EEGNet (suitable for small sample data) with Long Short-Term Memory (LSTM) to extract both spatial and temporal features, improving the classification accuracy [19]. Researchers from India introduced a network based on sliding windows and self-attention mechanisms to extract enhanced features from multi-segment EEG data, achieving better results [20]. Scholars from different regions have proposed transfer learning methods based on parameters and examples [21], [22] to address this issue with satisfactory results.

In summary, current methods have shown good results in decoding motor imagery signals, with increasing accuracy. However, the feature extraction prediction methods are still unable to identify unified representations of multiple motor imagery signals, leading to unsatisfactory classification results. Although deep learning models have continuously improved in prediction accuracy, their classification performance remains limited and requires further improvement, particularly in cross-subject prediction.

We propose a multi-branch self-attention network-based method for classifying motor imagery EEG signals in this study. Before the EEGNet fusion model, we decompose the signals into multiple frequency bands for the analysis of the EEG data of each subject. The decomposed signals are then fed into the EEGNet network to extract spatial features. The Self-attention block is used to highlight the significant features, and finally, an LSTM network is employed to extract more effective temporal features. This model utilizes multi-branch and self-attention mechanism to improve the accuracy of decoding motor imagery signals. In the motor imagery classification experiments, the model achieves accuracy of 84.04% and 71.67% for the subject-dependent and subject-independent. In addition, this paper compares and discusses the contribution of different signal decomposition methods to the accuracy of the model, ultimately selecting the most optimal EEG signal decomposition method for the proposed model.

The main contributions of this paper are summarized as follows:

- We propose an end-to-end multi-branch deep learning model that incorporates EEGNet, self-attention mechanisms, LSTM, and various signal decomposition methods. This model effectively extracts features from EEG signals in different frequency bands, making it more interpretable and improving its generalization performance.
- We conduct experiments on various methods of detecting EEG signal, such as ensemble empirical mode decomposition (EEMD), wavelet packet decomposition (WPD), and frequency decomposition based on brain rhythms, comparing and discussing their contributions to the accuracy of the model.
- Our proposed model demonstrates superior performance compared to other models in both subject-dependent and subject-independent tests. Additionally, the features extracted from EEG signals across different frequency bands can be further analyzed, and alternative methods can be explored to enhance the model generalization capability, leading to improvements in subject-independent accuracy.

The remainder of this paper is organized as follows. Section II explains the input data format and the core modules of the model. Section III introduces the dataset, evaluation criteria, and experimental setup and results. Finally, Section IV concludes this paper.

II. PROPOSED MODEL

This section proposes a deep learning model that combines multi-branch architecture with self-attention network (Fig. 1) to classify different types of motor imagery EEG signals. As shown in Fig. 2, each branch of our model consists of four main blocks: signal decomposition block, convolutional block, attention block and LSTM block.

The signal decomposition block decomposes the signals into different frequency components, which are then passed through the convolution block to output spatiotemporal feature sequences. The self-attention block reorders the importance of the information in the spatiotemporal sequences using a multi-head self-attention(MSA), and concatenates the spatiotemporal sequences of different frequency signals. Finally, LSTM is used to extract the temporal features of the EEG signals, and a SoftMax classifier is employed to input them into a fully connected (FC) layer for output.

The convolution block extracts the spatiotemporal features of the signals, the self-attention block applies a weighting on the extracted features, the LSTM block extracts high-level temporal features, and the frequency decomposition of the signal effectively enhances the data, improving the model and

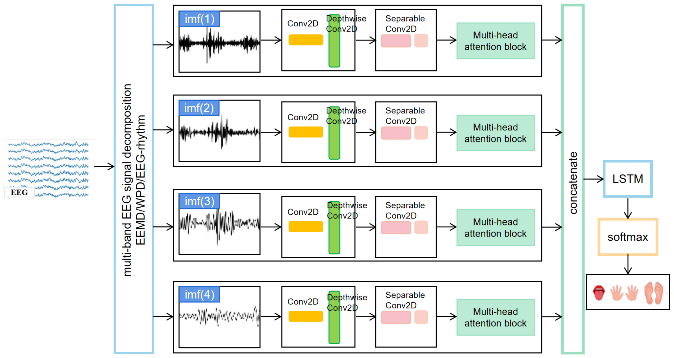

Fig. 1. The Constructed Multi-Branch Self-Attention Model

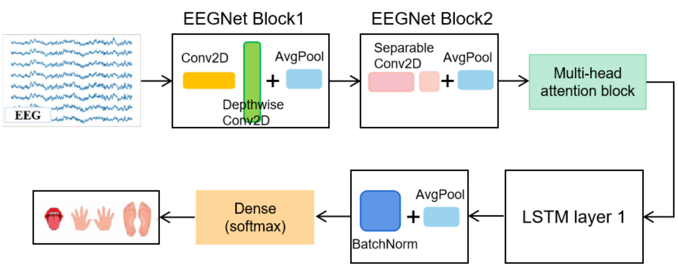

Fig. 2. The branch of the model

the interpretability of the extracted features, which in turn increases the accuracy of classification results. The following content describes the specific information of each block in the model.

A. Preprocessing and Model Input

In this study, we preprocess the raw signals and decompose them using different signal decomposition methods, then input the decomposed signals into the proposed model.

For the BCI competition IV 2a dataset [18], the data consists of C channels (EEG electrodes) and T time points (T = 1125 time points, C = 22 EEG channels), with labels corresponding to four motor imagery actions (left hand, right hand, foot, and tongue) represented as one-hot encoded four-dimensional vectors: [1,0,0,0], [0,1,0,0], [0,0,1,0], [0,0,0,1]. Each sample is selected for a duration of 4 seconds. Among the 7.5 seconds in each sample, only the signals recorded between 2 and 6 seconds are considered, during which a motor imagery cue is displayed on the screen, and the subject performs the task. Finally, the data is normalized.

B. Convolution Block

The design of the convolution block is mainly inspired by the EEGNet model proposed in [15], which uses deep convolutions and separable convolutions to extract and classify signal features.

The convolutional block comprises two parts. First, raw EEG signals are processed through a 2D convolutional layer (Conv2D) followed by depthwise separable convolutions (DepthwiseConv2D) to model inter-channel correlations. Subsequent average pooling reduces spatial dimensionality, while batch normalization and dropout (rate=0.5) mitigate overfitting. The second part employs separable convolutions to further distill channel-specific spatial features, followed by identical pooling and regularization operations. This hierarchical design generates compact spatiotemporal feature maps, balancing model capacity and computational efficiency.

In this study, different frequency signals obtained from decomposition use distinct parameters. In the first part, each branch uses different kernel sizes (8, 16, 32, 32) and filter numbers (1, 32), (1, 64), (1, 128), and (1, 128). The number of channels is set to the 22 channels specified in the experiment. Therefore, the filter size in DepthwiseConv2D is set to (22, 1). In the second part, separable convolution layers with a kernel size of 16 are used, with filters of (1, 8), (1, 16), (1, 32), and (1, 32).

C. Self-Attention Block

The self-attention block employs the multi-head self-attention (MSA) mechanism [23], which is a deep learning structure for processing sequential data. The MSA module consists of several self-attention layers with heads, where each self-attention layer has three main components: Query (Q), Key (K), and Value (V). The dot product between Q and K is computed, scaled, and passed through a SoftMax function to obtain attention weights, learning how much focus each position should have on others. The attention weights are then used to weight the values, resulting in the weighted representation for each position. This process is performed in parallel across multiple attention heads, with each head learning different aspects of the sequence. Finally, the outputs from all attention heads are concatenated and linearly mapped to produce the final output of the multi-head self-attention module. Adding the multi-head self-attention module allows the model to learn the relationships within the sequence, improving its ability to analyze global information.

In this study, the multi-head self-attention mechanism is applied to rank the importance of the spatiotemporal feature sequences output by the convolution block. The size of each attention head in the MSA is set to 8.

D. Long Short-Term Memory

In this study, Long Short-Term Memory (LSTM) networks are used to extract high-level temporal features of motor imagery signals, mainly because LSTM can avoid the vanishing gradient problem when processing long data sequences [24]. LSTM utilizes memory cells to store temporal information, which can retain information over long periods. These memory cells include three types of gates: the input gate, forget gate, and output gate. The state of the memory cell is updated at each time step to extract temporal features from the signal [25].

E. Signal Decomposition Block

This study utilizes three signal decomposition methods for decomposing motor imagery EEG signals: Empirical Mode Decomposition (EMD), Wavelet Packet Decomposition (WPD), and decomposition based on brain rhythms.

The study compares these three methods and discusses the most suitable one for signal decomposition. The three decomposition methods are as follows:

1) Empirical Mode Decomposition: Empirical Mode Decomposition(EMD) is suitable for the decomposition of nonlinear and non-stationary signals, breaking the signal into a series of Intrinsic Mode Functions (IMFs) that have good local properties and time-frequency characteristics. EEMD is a noise-assisted method based on EMD, which can reduce the impact of noise on decomposition results. It adds different white noise to the original signal and decomposes it multiple times to reduce the influence of noise, improving decomposition accuracy. The primary IMF signals (IMF1-IMF4) are used as the input to the model, which can help the model focus on different aspects of the data, aiding in the decoding of motor imagery signals.

2) Wavelet Packet Decomposition: Wavelet Packet Decomposition (WPD) is an extension of wavelet transform that allows signals to be decomposed at finer frequency levels. Compared to traditional wavelet transform, WPD is more flexible as it iteratively divides the signal into multiple sub-frequency bands, forming a tree-like structure. This decomposition captures local features of the signal more precisely, providing more frequency information. In this study, WPD is used to decompose motor imagery signals into low and high-frequency components, extracting both time-domain and frequency-domain features.

3) Decomposition Based on Brain Rhythms: This method decomposes EEG signals according to five brain rhythms: delta (0.5-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), beta (12-30 Hz), and gamma (30-80 Hz). Different frequency bands are associated with different cognitive and brain activities. For instance, alpha waves are typically associated with relaxation and wakefulness, theta waves with learning and creative thinking, and beta waves with focus and alertness. In motor imagery classification, focusing on specific frequency bands may be more important than others. This method involves bandpass filtering the signal into different frequency bands and extracting features from each band, making the output of the model more interpretable. A third-order Butterworth filter is used to decompose the EEG signal into four sub-frequency bands, with the selected cutoff frequencies being (0.5-8 Hz), (8-13 Hz), (12-30 Hz), (30-40 Hz). These sub-band signals are then passed through the model for further optimization of the optimal cutoff frequencies.

III. Experimental Results and Discussion

A. Dataset Selection and Evaluation Methods

1) Dataset Introduction: In this study, the BCI IV Competition 2a dataset[18] was chosen, which was registered and publicly provided by Graz University of Technology in Austria. The dataset consists of EEG data from 9 subjects. During the experiment, each subject was required to perform four distinct motor imagery tasks: left hand (class 1), right hand (class 2), both feet (class 3), and tongue (class 4). Each subject completed two sessions on different dates, with one session used for classifier training (T) and the other for testing (E). Each session was subdivided into 6 runs, with 48 trials per run, for a total of 288 tests per session (6 × 48). In each trial, 12 trials were allocated to each of the four motor imagery tasks, randomly ordered.

The EEG signals in this dataset are recorded from 25 electrode channels, 22 of which are EEG channels and 3 are EOG channels. Data from the 3 EOG channels are not used for classification. The signals are sampled at 250 Hz (250 samples per second) and bandpass filtered between 0.5 Hz and 100 Hz. The amplifier sensitivity is set to 100 μV. An additional 50 Hz notch filter is enabled to suppress line noise. The electrode configuration is shown in Fig. 3 (with the left image showing the locations of the EEG electrode and the right image showing the locations of the EOG electrode).

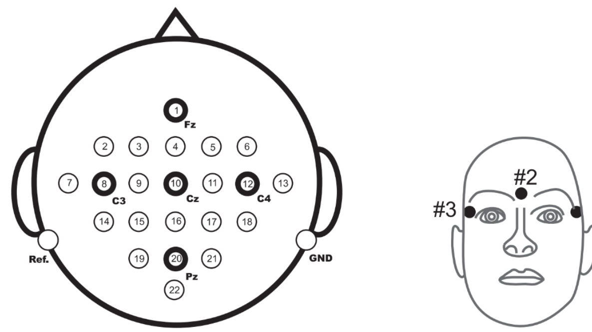

Fig. 3. Electrode Distribution of the BCI IV Competition 2a Dataset

2) Evaluation Methods: We evaluate model performance under two paradigms: (1) Subject-dependent testing: Data from two sessions per subject are partitioned into training (Session T) and testing (Session E) sets. (2) Subject-independent testing: Leave-one-subject-out cross-validation (LOSO-CV) is adopted, where models are trained on data from eight subjects and validated on the excluded subject. This rigorous setup assesses both personalized adaptation and cross-subject generalization capabilities.

3) Training Parameters: After multiple experiments, the same configuration was used across all tests. The model was implemented in PyCharm, using the Adam optimizer with a learning rate of 0.0001, a batch size of 64, and an early stopping strategy. The maximum number of epochs was set to 1000, with a patience of 300 epochs for early stopping.

B. Comparison with Recent Studies

The proposed model utilizes a multi-branch attention mechanism for motor imagery classification. Three different signal decomposition methods were tested and the accuracy results were compared with several classic motor imagery models. The accuracy and standard deviation of the models were analyzed.

1) Subject-Dependent & Cross-Session Results: To compare the proposed model, EEGNet and EEGTCNet were used as baselines. Additionally, the multi-band models were compared with FBCSP, Deep ConvNet, Shallow ConvNet, and TCNet_Fusion, with the results shown in TABLE I. Each entry in the table represents the percentage accuracy of a

TABLE I

THE SUBJECT-DEPENDENT AND CROSS-SESSION EXPERIMENTAL RESULTS FOR THE BCI IV COMPETITION 2A DATASET

	FBCSP	Deep ConvNet	Shallow ConvNet	EEGNet	EEGTCNet	TCNet_Fusion	WPD	EEMD	Multi-Band
A1	65.6	77.8	79.5	85.6	85.8	86.1	86.7	86.1	**89.6**
A2	56.9	56.1	56.3	49.6	56.0	55.8	62.6	60.2	**65.6**
A3	65.3	84.7	88.9	90.7	89.7	88.9	91.6	88.9	**94.8**
A4	62.2	60.4	**80.9**	67.1	70.9	70.9	64.5	74.3	69.1
A5	26.0	66.8	57.3	62.0	65.7	66.3	70.8	71.8	**76.7**
A6	47.9	61.4	53.8	53.7	57.2	59.2	59.8	**66.7**	63.5
A7	75.4	83.7	91.7	82.3	90.1	90.1	89.4	90.3	**92.0**
A8	64.2	78.7	81.3	81.7	84.5	84.7	84.0	84.5	**86.1**
A9	62.2	80.2	79.2	**84.1**	83.1	82.3	83.5	80.3	84.1
Avg	58.32	72.17	74.31	72.97	75.87	76.03	76.98	78.12	**80.18**
Std	13.65	11.01	14.54	15.13	13.64	13.20	12.5	10.49	11.81

TABLE II

THE SUBJECT-DEPENDENT AND NON-CROSS-SESSION EXPERIMENTAL RESULTS FOR THE BCI IV COMPETITION 2A DATASET

	FBCSP	Deep ConvNet	Shallow ConvNet	EEGNet	EEGTCNet	TCNet_Fusion	WPD	EEMD	Multi-Band
A1	81.6	77.1	84.4	**88.5**	84.0	86.1	86.8	86.7	86.1
A2	52.8	56.1	62.2	66.0	66.3	66.0	65.5	66.9	**70.6**
A3	84.4	91.0	90.3	**95.1**	94.1	93.4	94.9	91.6	93.1
A4	65.3	74.3	74.0	73.6	72.6	72.6	67.1	76.8	**80.6**
A5	56.3	78.8	70.5	75.4	76.0	79.9	79.5	82.2	**84.4**
A6	44.4	68.4	56.9	64.2	62.9	66.7	64.1	**68.8**	68.7
A7	89.2	91.0	91.7	90.3	89.9	90.3	91.4	**94.2**	92.1
A8	81.9	81.3	84.1	85.8	84.7	85.8	87.9	88.9	**89.9**
A9	72.9	82.3	82.9	86.5	85.4	85.4	89.3	85.0	**90.8**
Avg	69.87	77.81	77.44	80.60	79.54	80.69	80.72	82.35	**84.04**
Std	15.90	10.93	12.28	11.11	10.68	10.06	12.08	9.68	9.06

given method for each subject in dataset 2a. Labels A1 to A9 correspond to the 9 subjects in the dataset, and the mean accuracy and standard deviation are reported.

The multi-band model achieved satisfactory results on the BCI IV 2a dataset, with the three signal decomposition methods reaching accuracy rates of 76.98%, 78.12%, and 80.18%, respectively. These results represent improvements of 4.01%, 5.15%, and 7.21% over the precision of the EEGNet baseline. This improvement is attributed to the multi-band signal decomposition, which extracts frequency-domain features more effectively. The attention mechanism further enhances feature discriminability.

TABLE I reveals that certain subjects performed better with EEMD signal decomposition (Subjects 4, 5, 6, 7, 8), while others showed better performance with WPD decomposition (Subjects 1, 2, 3, 9). This is likely due to the different features captured by each decomposition technique. Comparing the accuracy of other models, the multi-band model we proposed performs similarly to the other models.

2) Subject-Dependent & Non-Cross-Session Results: With the introduction of more complex network architectures in deep learning models, the interpretability of the models has also become more complicated. To further validate the stability of the model proposed in this study, experiments were conducted without cross-session validation, using ten-fold cross-validation on the dataset. The specific results are shown in TABLE II.

Upon observing the performance data of the multi-band model in TABLE II, the average accuracies of our proposed model reached 80.72%, 82.35%, and 84.04%, which represents an improvement of 0.12%, 1.75%, and 3.44% over the baseline EEGNet, respectively. Moreover, the standard deviations of the EEMD and brain rhythm decomposition methods were also smaller than those of EEGNet and other models, indicating less fluctuation between subjects compared to other models.

The tendencies of different subjects for different signal decomposition methods were similar to the results observed in cross-session testing, suggesting that subjects indeed show preferences for specific signal decomposition methods.

Overall, both in cross-session and non-cross-session scenarios, our model performs better than other models on the BCI IV 2a dataset. As for the choice of signal decomposition method, although EEMD decomposition in the multi-band model performs best for certain subjects (Subjects 6 and 7), the method using brain rhythm-based frequency decomposition shows the highest accuracy based on average accuracy and standard deviation. Therefore, we primarily choose the method of decomposing signals based on brain rhythms.

3) Subject-Independent Results: This paper also needs to investigate another aspect of the model performance, namely its generalization ability. Leave-one-out validation was employed for experiments, where the baseline model and our model were tested on the datasets, yielding the results shown in TABLE III.Through leave-one-out validation, it was found that in cross-subject testing, the model accuracy generally decreases by about 10%. This is because each subject's EEG signal features are quite unique, so previously useful features may lose their effectiveness in cross-subject testing, leading to a decrease in accuracy.

Upon observing the experimental results in TABLE III, it can be seen that for the EEMD and WPD signal decom-

TABLE III
THE SUBJECT-DEPENDENT AND SUBJECT-INDEPENDENT EXPERIMENTAL RESULTS FOR THE BCI IV COMPETITION 2A DATASET

Models	subject-dependent	subject-independent
FBCSP	69.87	/
Deep ConvNet	77.81	/
Shallow ConvNet	77.44	/
EEGNet	80.59	68.79
EEGTCNet	79.54	69.52
TCNet_Fusion	80.69	70.58
WPD EEG_LSTM	80.72	60.1
EEMD EEG_LSTM	82.35	66.92
Multi-Band EEG_LSTM	84.04	71.67

position methods, the accuracy in cross-subject scenarios is lower than that of the baseline model, with a significant drop. This is mainly because the decomposition results for each subject lack commonality, thus leading to poor performance in cross-subject testing. However, using brain rhythm-based signal decomposition (Multi-Band EEG_LSTM) in the model resulted in better accuracy in cross-subject testing, with a 2.88% improvement over EEGNet, and it performed comparably to other models in terms of cross-subject prediction accuracy. We believe that the similarity of features across different subjects in the same brain rhythm frequency band effectively improves the accuracy of subject-independent experiments. Therefore, choosing brain rhythm-based signal decomposition is more reasonable and also provides a certain degree of interpretability to the model.

4) Confusion Matrix Comparison: Fig. 4 shows the confusion matrix for the four motor imagery tasks (left hand, right hand, both feet, and tongue) in cross-session and subject-dependent settings. The matrix provides a comprehensive overview of classification performance. Our model

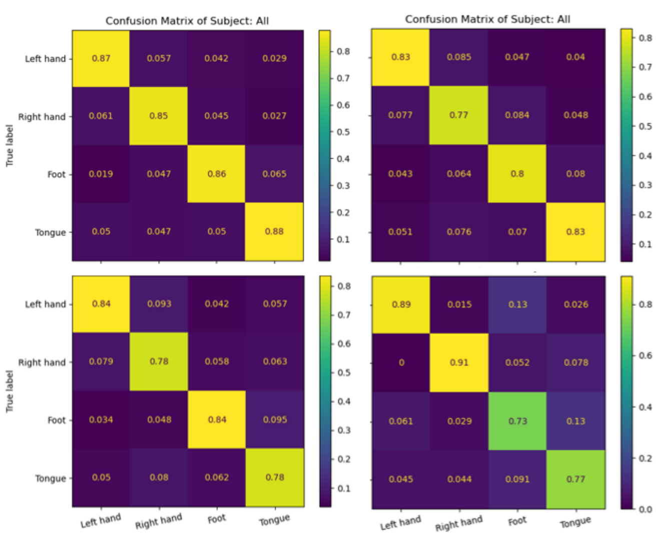

Fig. 4. Comparison of confusion matrices: Our Model (left 1), EEGNet (right 1), EEGTCNet (left 2), TCNet_Fusion (right 2)

outperforms others in classification tasks, especially for both feet and tongue movements. It demonstrated the best diagonal values (correct classifications) compared to other models, with the least off-diagonal errors (misclassifications).

Overall, while TCNet_Fusion excelled in recognizing left-hand & right-hand motor imagery, Our model showed superior classification performance across all tasks.

C. Comparison of Different Signal Decomposition Methods

This section will provide a discussion of signal decomposition techniques (EEMD, WPD, and Brain Rhythm Decomposition). This study compares the accuracy of three

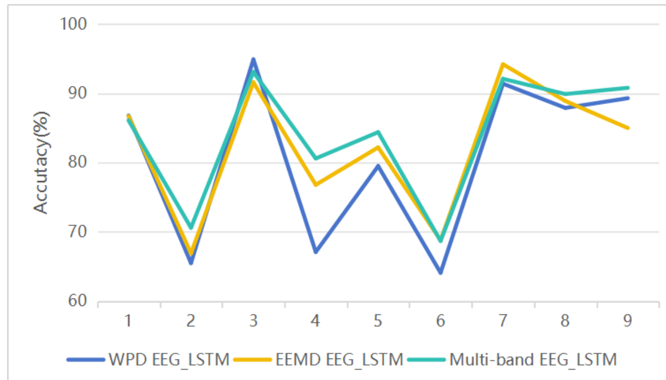

Fig. 5. The accuracy of different signal decomposition methods in our model.

signal decomposition methods, as shown in Fig. 5. From the Fig. 5, it can be seen that the accuracy varies across different subjects and decomposition methods, indicating that different subjects may be better suited to different signal decomposition techniques. However, overall, the signal decomposition based on brain rhythms shows a higher accuracy compared to the other two methods. In the case of non-cross-subject testing, the performance of WPD and EEMD decomposition methods significantly decreased, while the decomposition based on brain rhythms performed better. We believe the main reason is that the decompositions of WPD and EEMD are subject-specific, whereas the brain rhythm-based decomposition exhibits a certain degree of commonality across different subjects, which enhances the model's generalization ability. Therefore, our model ultimately chose brain rhythm-based signal decomposition.

IV. CONCLUSIONS

This paper proposes a multi-branch self-attention network to decode motor imagery EEG signals. In this approach, various signal decomposition methods are used to separate motor imagery signals into low-frequency and high-frequency components. These components are individually processed through convolutional modules to extract spatiotemporal feature sequences.

This design facilitates the model in capturing distinct frequency-specific features of motor imagery signals, enhancing interpretability. Additionally, the self-attention mechanism identifies and weights the most critical features within the spatiotemporal feature sequences, improving classification accuracy. Finally, an LSTM module extracts high-level temporal features, reflecting the dynamic characteristics of motor imagery signals. By combining these components, the proposed model achieves robust multi-dimensional feature recognition, demonstrating superior performance in decoding motor imagery signals.

This study also analyzes the contributions of individual modules to the overall model performance, particularly exploring the impact of the signal decomposition module. Three decomposition methods were compared: empirical mode decomposition (EEMD), wavelet packet decomposition (WPD), and brain rhythm-based decomposition. Among these, the brain rhythm-based method yielded the best results. Ultimately, the proposed model achieved accuracies of 84.04% and 71.67% for subject-dependent and subject-independent tests, respectively, surpassing selected baseline models. These advancements improve the recognition accuracy of motor imagery signals, with potential applications in rehabilitation and control within medical fields.

Future work will focus on improving generalization performance by exploring feature-based transfer learning. This would validate the representativeness of features extracted from different frequency bands post-decomposition, enhancing the model's interpretability and generalizability while reinforcing the contributions outlined in this study.

V. Ethics Statement

This study used the publicly available BCI Competition IV 2a dataset, for which the original investigators obtained ethical approval and informed consent. No further ethical approval was required as the data are fully anonymized and publicly accessible. The study complies with the Declaration of Helsinki as revised in 2000.

References

[1] I. Ahmed, G. Jeon, and F. Piccialli, "From artificial intelligence to explainable artificial intelligence in industry 4.0: A survey on what, how, and where," *IEEE Transactions on Industrial Informatics*, vol. 18, no. 8, pp. 5031–5042, 2022.

[2] L. Bougrain, M. Clerc, and F. Lotte, *Brain Computer Interfaces: Methods, Applications and Perspectives*. John Wiley & Sons, Incorporated, 2016.

[3] R. Abiri, S. Borhani, E. W. Sellers, Y. Jiang, and X. Zhao, "A comprehensive review of eeg-based brain–computer interface paradigms," *Journal of neural engineering*, vol. 16, no. 1, p. 011001, 2019.

[4] T. M. Vaughan, W. J. Heetderks, L. J. Trejo, W. Z. Rymer, M. Weinrich, M. M. Moore, A. Kübler, B. H. Dobkin, N. Birbaumer, E. Donchin *et al.*, "Brain-computer interface technology: a review of the second international meeting." *IEEE transactions on neural systems and rehabilitation engineering: a publication of the IEEE Engineering in Medicine and Biology Society*, vol. 11, no. 2, pp. 94–109, 2003.

[5] J. Klonovs, C. K. Petersen, H. Olesen, and A. Hammershoj, "Id proof on the go: Development of a mobile eeg-based biometric authentication system," *IEEE Vehicular Technology Magazine*, vol. 8, no. 1, pp. 81–89, 2013.

[6] B. Blankertz, R. Tomioka, S. Lemm, M. Kawanabe, and K.-R. Muller, "Optimizing spatial filters for robust eeg single-trial analysis," *IEEE Signal processing magazine*, vol. 25, no. 1, pp. 41–56, 2007.

[7] S. Kumar, A. Sharma, and T. Tsunoda, "An improved discriminative filter bank selection approach for motor imagery eeg signal classification using mutual information," *BMC bioinformatics*, vol. 18, pp. 125–137, 2017.

[8] C. Vidaurre, M. Kawanabe, P. von Bünau, B. Blankertz, and K.-R. Müller, "Toward unsupervised adaptation of lda for brain–computer interfaces," *IEEE Transactions on Biomedical Engineering*, vol. 58, no. 3, pp. 587–597, 2010.

[9] Y. Li and C. Guan, "An extended em algorithm for joint feature extraction and classification in brain-computer interfaces," *Neural Computation*, vol. 18, no. 11, pp. 2730–2761, 2006.

[10] S. Sun and C. Zhang, "Adaptive feature extraction for eeg signal classification," *Medical and Biological Engineering and Computing*, vol. 44, pp. 931–935, 2006.

[11] N. Lu, T. Li, X. Ren, and H. Miao, "A deep learning scheme for motor imagery classification based on restricted boltzmann machines," *IEEE transactions on neural systems and rehabilitation engineering*, vol. 25, no. 6, pp. 566–576, 2016.

[12] S. Sakhavi, C. Guan, and S. Yan, "Learning temporal information for brain-computer interface using convolutional neural networks," *IEEE transactions on neural networks and learning systems*, vol. 29, no. 11, pp. 5619–5629, 2018.

[13] ——, "Parallel convolutional-linear neural network for motor imagery classification," in *2015 23rd European signal processing conference (EUSIPCO)*. IEEE, 2015, pp. 2736–2740.

[14] Y. R. Tabar and U. Halici, "A novel deep learning approach for classification of eeg motor imagery signals," *Journal of neural engineering*, vol. 14, no. 1, p. 016003, 2016.

[15] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "Eegnet: a compact convolutional neural network for eeg-based brain–computer interfaces," *Journal of neural engineering*, vol. 15, no. 5, p. 056013, 2018.

[16] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," *arXiv preprint arXiv:1803.01271*, 2018.

[17] T. M. Ingolfsson, M. Hersche, X. Wang, N. Kobayashi, L. Cavigelli, and L. Benini, "Eeg-tcnet: An accurate temporal convolutional network for embedded motor-imagery brain–machine interfaces," in *2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. IEEE, 2020, pp. 2958–2965.

[18] C. Brunner, R. Leeb, G. Müller-Putz, A. Schlögl, and G. Pfurtscheller, "Bci competition 2008–graz data set a," *Institute for knowledge discovery (laboratory of brain-computer interfaces), Graz University of Technology*, vol. 16, pp. 1–6, 2008.

[19] I. H. de Oliveira and A. C. Rodrigues, "Empirical comparison of deep learning methods for eeg decoding," *Frontiers in Neuroscience*, vol. 16, p. 1003984, 2023.

[20] H. Altaheri, G. Muhammad, and M. Alsulaiman, "Physics-informed attention temporal convolutional network for eeg-based motor imagery classification," *IEEE transactions on industrial informatics*, vol. 19, no. 2, pp. 2249–2258, 2022.

[21] R. Zhang, Q. Zong, L. Dou, X. Zhao, Y. Tang, and Z. Li, "Hybrid deep neural network using transfer learning for eeg motor imagery decoding," *Biomedical Signal Processing and Control*, vol. 63, p. 102144, 2021.

[22] K. Zhang, G. Xu, L. Chen, P. Tian, C. Han, S. Zhang, and N. Duan, "Instance transfer subject-dependent strategy for motor imagery signal classification using deep convolutional neural networks," *Computational and Mathematical Methods in Medicine*, vol. 2020, no. 1, p. 1683013, 2020.

[23] A. Vaswani, "Attention is all you need," *Advances in Neural Information Processing Systems*, 2017.

[24] S. Hochreiter, "Long short-term memory," *Neural Computation MIT-Press*, 1997.

[25] J. Du, C.-M. Vong, and C. P. Chen, "Novel efficient rnn and lstm-like architectures: Recurrent and gated broad learning systems and their applications for text classification," *IEEE transactions on cybernetics*, vol. 51, no. 3, pp. 1586–1597, 2020.